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Probabilistic Combinatorics SS24

Sheet 5

Due: 20. May at 19:00

(1) Show that for the Erdős-Rényi model, if$p_n \ll 1/n$, then

$$\mathbb{P}(G(n,p_n)\text{contains a cycle}) \to 0.$$

Here, a cycle (of length k) in a graph $G=(V,E)$ a sequence of k pairwise distinct vertices $v_1,...,v_k \in V$, such that $\{v_i,v_{i+1}\} \in E$ for $1 \leq i < k$ and $\{v_k,v_1\} \in E$.

(2) (i) Show that for $n \geq 3$ and $G=(V,E)$ a graph with $\#V = n$ and $\#E \geq n$, there is a cycle in G.

(ii) Show (using 1), that $p(n) = 1/n$ is a threshold function for the monotone graph property "G(n,p) has a cycle".

(3) Show for $\mathcal{P} = \{\mathcal{P}_n\}_{n\in\mathbb{N}}$ eine monotone graph property and parameters $0 \leq p_1 \leq p_2 \leq 1$ that:

$$\mathbb{P}(G(n,p_1) \in \mathcal{P}) \leq \mathbb{P}(G(n,p_2) \in \mathcal{P}).$$

Here $\mathbb{P}$ denotes the probability induced by the corresponding parameter in $G(n,p)$.

Hinweis: Think of $G(n,p_2)$ as a two-stage probability experiment.

(4) Let $\Omega_{n,k} = \{G=(V,E) \mid V=[n],\ \#E=k\}$ and $\Omega_n = \{G=(V,E) \mid V=[n]\}$ the set of graphs on $[n]$ with k edges or with an arbitrary number of edges respectively. We define the *Erdös-Renyi-Model with edge number k* as the probability space $(\Omega_{n,k}, \mathbb{P}_{n,k})$ with $\mathbb{P}_{n,k}$ the uniform distribution.

A map $X : \Omega_n \to \mathbb{R}$ can (since the probability spaces are discrete) be thought of as a random variable on $G(n,p)$. Hence for every $0 \leq p \leq 1$ there is an expected value $\mathbb{E}_p[X]$. Furthermore, the restriction of X on $\Omega_{n,k}$ can be considered; this is a random variable on $\Omega_{n,k}$. Show under this identification:

$$\sum_{k=0}^{\binom{n}{2}} \mathbb{E}[X_{|\Omega_{n,k}}] = \left(\binom{n}{2}+1\right) \int_0^1 \mathbb{E}_p[X]\ \mathrm{d}p.$$